

# Abdelhak Bourdim

## Senior Embedded Software Engineer — C/C++ • RTOS • Embedded Linux

**Date of Birth:** 20 August 1969      **Nationality:** French      **Marital Status:** Married      **Visa:** Requires employer-sponsored work visa  
**Email:** abdelhak.bourdim@gmail.com      **Phone:** +33 6 19 51 51 73      **Location:** Paris area, France  
**LinkedIn:** linkedin.com/in/abdelhak-bourdim-96416122      **Website:** workshop-diy.org

Mobility: UAE, Saudi Arabia, Qatar — Available immediately

Native Arabic Speaker — Key Advantage for the GCC Market

## PROFILE

Embedded software specialist with **over 20 years of experience** developing firmware, middleware, and application layers — bare metal, RTOS, and Embedded Linux — for critical and industrial systems. Designing low-level drivers, communication stacks (CAN, BLE, USB, TCP/IP), and embedded applications on ARM Cortex, STM32, Microchip, and NXP architectures.

Delivered certified solutions for **aerospace** (Safran, Thales — DO-178), **medical devices** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT** (**cybersecurity** (Enedis — Linky), **payment systems** (Cartadis), and **V2G smart charging** (E2-CAD — ISO 15118). MISRA C compliant development. **Trilingual:** French, English, and Native Arabic.

## PROFESSIONAL EXPERIENCE

### E2-CAD — Eragny, France

May 2025 – Mar. 2026

#### Embedded Software Engineer — EV Charging / V2G

EV Charging Station — Iotecha IP

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), debug tools setup (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, Saleae)
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Charging station debug and commissioning (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, CAN, TCP/IP, Saleae, HackRF One, Python, Bash

### Safran — Creteil, France

Mar 2024 – May 2025

#### Embedded Software Engineer — Aerospace (SSPC)

- Development and update of low-level C drivers for the SSPC system
- Implementation of logging tools for diagnostics and exchange traceability
- Writing and execution of on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

### Asterios Technologies — Massy, France

Aug 2023 – Feb 2024

#### Embedded Software Engineer — Aerospace (SOM Testing)

- C development of validation tests for the SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing compliant with project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polarion

### Thales — Gennevilliers, France

May 2022 – Jul 2023

#### Embedded Software Engineer — Defense Radio

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- Development and execution of HLT (High Level Tests)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

### Valeo — Creteil, France / Egypt

Mar – Apr 2022

#### Embedded Software Engineer — Test Audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation proposals and client/system requirements review

Env: NI Mastre, Vector CANoe

### Arkamys — Levallois-Perret, France

Jul 2021 – Feb 2022

#### Embedded Software Engineer — Automotive Audio

- GStreamer plugin and CAN driver (ELM327) development, diagnostic tools
- Unit and integration testing (Valgrind), build system Buildroot

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

**Thales — Velizy-Villacoublay, France**

Nov 2020 – Jun 2021

**Embedded Software Engineer — Autonomous Train (ATO)**

- Application migration from 32-bit to 64-bit on Embedded Linux platform
- IPC communication protocol definition and implementation
- Network diagnostic tools development and integration testing

**Env:** Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Python, Jira**PK Vitality — Paris, France**

Jun – Sep 2020

**Embedded Software Engineer — Medical Device (CGM)**

- HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- STM32 TouchGFX touchscreen, FreeRTOS, low power mode
- In vitro testing and Doxygen documentation (IEC 62304 compliant)

**Env:** STM32L459, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen**Zodiac Aerospace (Safran) — Montreuil / Vancouver, Canada**

Feb 2017 – Mar 2020

**Embedded Software Engineer — Aerospace (SSPC A350)**

- Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- uAFDX to CAN gateway implementation and diagnostic functions
- SSPC A350 application layer development and optimization
- Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

**Env:** Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C**Enedis — Nanterre, France**

May 2015 – Sep 2016

**Embedded Cybersecurity Engineer — Smart Grid**

- Linky concentrator security diagnostics and audit
- Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)

**Env:** Kali Linux, Embedded Linux, Nmap, Wireshark, Scapy, SDR, Python, Bash**Sorin — Clamart, France**

Nov 2014 – Apr 2015

**Embedded Software Engineer — Medical (BLE Pacemaker)**

- Bluetooth Smart module validation, Serial Port Service Profile development
- Smartphone communication protocol development and Android application

**Env:** Cortex M0, Keil, C, BLE, Android**General Electric Healthcare — Buc / USA**

2012 – 2014

**Embedded Software Engineer — Medical Imaging (X-Ray)**

- CANOpen stack integration (slave/master), CAN driver and service development
- CMX-RTX kernel porting, USA team technical support and on-site visits

**Env:** TI TMS320, ARM Cortex A8, VxWorks, C, C++**Photomaton / BCM / Vertical M2M / PGS — Paris**

2010 – 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

**Cartadis — Fontenay-sous-Bois, France**

Oct 2001 – Dec 2009

**8 years** — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)**TECHNICAL SKILLS**

|                    |                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------|
| Languages          | C, C++, Python, Bash, Assembly, Perl, Awk                                                   |
| OS / RTOS          | Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel                                         |
| Microcontrollers   | ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto |
| Protocols / Bus    | CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP               |
| Stacks             | TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer                                  |
| Security / Payment | Smart Card, RFID Mifare, DES, RC4, USB CCID, embedded pentesting                            |
| Tools              | Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX  |
| Management         | Git, Synergy, Jira, Polarion, Doxygen, Wireshark, Saleae                                    |

**STANDARDS & METHODS**

|        |         |           |                 |                        |                  |      |         |               |
|--------|---------|-----------|-----------------|------------------------|------------------|------|---------|---------------|
| DO-178 | MISRA C | IEC 62304 | ISO 15118 (V2G) | ISO 14443 (Smart Card) | ISO 15693 (RFID) | OCPP | V-Model | Agile / Scrum |
|--------|---------|-----------|-----------------|------------------------|------------------|------|---------|---------------|

**EDUCATION****AFPA, Champs-sur-Marne, France**

Engineer — Electronic and Computer Systems Design

**Tenon Hospital, Paris, France**

Postgraduate Diploma — Biomedical Instrumentation

**INDUSTRY SECTORS**

Aerospace (Safran, Thales, Zodiac)

Medical (GE Healthcare, Sorin, PK Vitality)

Automotive (Valeo, Arkamys)

Energy / Smart Grid (Enedis — Linky)

## LANGUAGES

|         |                                    |
|---------|------------------------------------|
| French  | Native / Bilingual                 |
| English | Professional working               |
| Arabic  | Native / Bilingual — KEY ADVANTAGE |

EV Charging / V2G (E2-CAD)  
IoT / M2M (Vertical M2M)  
Payment (Cartadis, Photomaton, BCM)

## VOLUNTEER WORK

### Workshop-DIY

Founder — Chelles, France — since 2020

Robotics & STEM workshops for youth (ages 6-12+)

ESP32 robot kit design (WiFi, BLE, servos, sensors)

18+ workshops: Arduino, micro:bit, AI, 3D printing

[workshop-diy.org](https://workshop-diy.org)

References available upon request